Embedded Systems Development

Exercise 3

1. Design a "click counter" using an ATmega328P microcontroller and present your design as a schematic diagram.

   The "click counter" should count the number of times a pushbutton switch is pressed and display the count in binary using LEDs. Initially, it should display 0 on the LEDs by turning all LEDs off. The user can then enter a number by repeatedly pressing a pushbutton switch. When the user indicates that the number entry is done by pressing a second pushbutton switch, the "click counter" should display the entered number in binary on the LEDs. The displayed number should remain on the LEDs until the second pushbutton is pressed again, at which point the display should be cleared by turning all LEDs off. The "click counter" should be capable of counting and displaying any number from 0 to 255.

   As the pins 2 and 3 of ATmega328P is multiplexed for receiving and transmitting data over a serial communication, they cannot be used as General Purpose Input/Output (GPIO) pins while the microcontroller serially communicates. Therefore, reserve these pins in your design, as they will be used for serial communication in a future extension.

2. Write the required firmware for the "click counter" in C.